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Introduction

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Introduction: what is a tipping point, its dynamics and importance

Chapter 1

One-dimensional tipping-point techniques

THIS IS COMPLETE AS FAR AS I'M CONCERNED. I AM INTERESTED IN YOUR FEEDBACK.

There exist many methods for the detection and prediction of tipping points in dynamical systems Held & Kleinen GRL 2004 [Ditlevsen and Johnsen, 2010; Lenton et al., 2012; Ashwin et al., 2012; Livina et al., 2012] and, whilst many dynamical systems, including almost all examples of physical systems studied in ecology, geosciences, climate sciences, etc. are ~~extremely~~ high-dimensional, they often have only one trajectory variable, such as temperature, ~~we~~ concentrate in this chapter on methods for detecting tipping points in one-dimensional systems. Specifically of interest here are methods which provide an early warning signal (EWS) of the tipping event. Typically, an increase in variance may be used as an EWS as may autocorrelation [Carpenter and Brock, 2006; Dakos et al., 2008].

Central to the methods presented in this chapter is the concept of critical slowing down. As a system approaches a tipping point, such as a bifurcation, in which a stable mode is lost, it will take longer to return to that decaying stable mode, known as a "critical mode" after any perturbation [Ashwin et al., 2012]. This effect suggests an increase in autocorrelation as the dynamics change from something like white noise around the stable node, to a progression away from it.

Not only does the concept of critical slowing down suggest that the autocorrelation increases, but also the autocorrelation scaling exponent changes since the autocorrelation function (ACF) will experience different rates of change depending on the lag. Other, related, time-scaling properties of a system could also provide an EWS for tipping points. For example, spectral properties of time series are suggested to predict tipping points [Kleinen et al., 2003] and the detrended fluctuation analysis (DFA) exponent, which is related to the power spectrum scaling exponent [Heneghan and McDarby, 2000] and the autocorrelation scaling exponent [Kantelhardt et al., 2001], has been used in various tipping point applications [Livina and Lenton, 2007; Lenton et al., 2012]. Intuitively, we can understand that the time-scaling properties of a system in a stable state will be different to that undergoing a tipping event since, by its nature, a tipping event will not scale.

do not?

We do here examine methods to detect the modes of a generating dynamical system, such as the potential analysis method [Livina et al., 2011, 2012], but methods designed to predict an impending tipping point, known as Early Warning Signals. In this Chapter we present three scaling exponents in section 1.1 and explore the relationships between them in section 1.2. In section 1.3 we then provide examples of the use of these exponents as early warning signals applied to time series.

1.1 Time series scaling exponents

Scaling properties of time series can be measured using three techniques: the autocorrelation function (ACF), detrended fluctuation analysis (DFA) and the power spectrum [Bak et al., 1988; Kantelhardt et al., 2001]. In this section we will describe the three methods, in the next section, consider their relationships to each other. In the following sections we will demonstrate the use of these exponents as Early Warning Signals when applied to time series data from simple dynamical systems. The DFA exponent has previously been utilised in a method to provide as

EWS [Livina and Lenton, 2007], but the use of the power spectrum scaling exponent in this way is novel [Prettyman et al., 2018].

1.1.1 The autocorrelation exponent

The autocorrelation function (ACF) of a time series, with lag l, is estimated using the formula,

std usually is denoted σ

$$\mathrm{ACF}_l(X) = \frac{1}{(N-l)s^2}\sum_{j=1}^{N-l}(X_j-\bar{X})(X_{j+l}-\bar{X}) \tag{1.1}$$

where $\bar{X}$ and s are the mean and standard deviation of the series X. The ACF scaling exponent γ is defined as the power-law decay rate of the autocorrelation function with increasing lag [Kantelhardt et al., 2001], that is,

$$ACF_l \sim l^{-\gamma}, \tag{1.2}$$

for long-range correlations. We are therefore able to estimate the exponent γ by calculating the ACF with a range of lags $l = 1,2,3,...$ and plotting the results ~~plotted~~ on logarithmic axes. γ can then be found by calculating the negative slope of the linear fit to the points on this plot. In figure 1.1a, a long-range correlated red noise signal is plotted, this noise is generated using the method presented by Kasdin [1995] (implemented in MatLab's DSP system toolbox). Simply, the method uses an auto-regressive model of order 63 (an AR63 model) with a zero-mean white noise signal $\eta(t)$:

$$\sum_{i=0}^{63} a_i z(t-i) = \eta(t), \tag{1.3}$$

where the AR coefficients a_i are calculated recursively

$$a_0 = 1 \quad a_k = \left(k-1-\frac{\lambda}{2}\right)\frac{a_{k-1}}{k}\; k = 1,2,.... \tag{1.4}$$

We use a power spectral density exponent $\lambda = 2$ to produce the resulting red noise signal $z(t)$.

In figure 1.1b the exponent γ of the red noise signal is calculated by measuring the value of the negative slope of the curve plotted in logarithmic axes. In this example the slope is calculated in the range $10 \le l \le 100$ which agrees with the range used by Livina and Lenton [2007] in the measurement of the DFA exponent.

Typically the simple lag-1 ACF, calculated using equation 1.1 with $l = 1$, is used in early warning signal contexts, including in this thesis, rather than the ACF scaling exponent. This is due to the ACF exponent having large variability in comparison to the DFA exponent, this and other effects are examined in section 1.2. Other measures of autocorrelation, such as the Mann-Kendall coefficient, may be used [Yue et al., 2002], but are not examined here.

1.1.2 Detrended Fluctuation Analysis

Detrended Fluctuation Analysis (DFA), defined by Kantelhardt et al. [2001] is used by Livina and Lenton [2007] as an EWS, similarly to ~~auto~~correlation. We therefore find it appropriate to draw comparisons between ~~the~~ two methods. Detrended Fluctuation Analysis has been applied successfully in various contexts [Peng et al., 1994, 1995; Koscielny-Bunde et al., 1998; Bunde et al., 2000]. Here we describe the method used to find the DFA exponent α.

Taking time series data $z(t)$ of length N, the DFA method involves calculating the cumulative sum of $z(t)$, which is called Y. This cumulative series is then split this into $\lfloor N/s \rfloor$ non-overlapping segments $Y_{(j)}$ and, for each segment, a 'detrended' series, $\overline{Y}_{(j)}$, is found by subtracting an order-n polynomial fit $p_n(Y_{(j)})$. Equations 1.5 to 1.7 ? unfinished sentence

$$y_i = \sum_{k=0}^{i} x_k \quad i = 0, \ldots, N \tag{1.5}$$

$$Y_{(j)} = \{y_{js}, y_{js+1}, \ldots, y_{(j+1)s-1}\} \quad j = 0, 1, \ldots, \left\lfloor \frac{N}{s} \right\rfloor - 1 \tag{1.6}$$

$$\overline{Y}_{(j)} = Y_{(j)} - p_n(Y_{(j)}) \quad j = 0, 1, \ldots \tag{1.7}$$

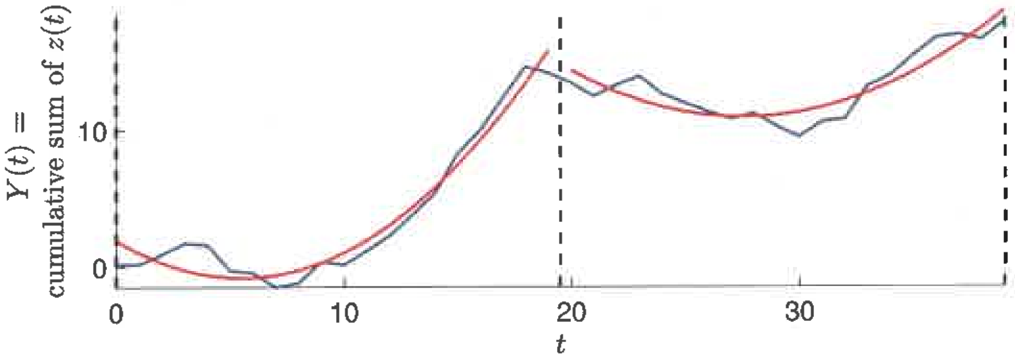

Fig. 1.2 Showing the detrending step in the order-2 DFA algorithm (equation 1.7). The cumulative sum of a pink-noise time series $z(t)$ is shown (blue line) with the quadratic best fit (red line) in each segment of length 20 (marked by dashed vertical lines).

Figure 1.2 illustrates the detrending step of the DFA algorithm: the time series $z(t)$ is pink-noise series of length 40 generated using equation 1.3 with parameter $\lambda = 1$ and the method shown is order-2 DFA, where a quadratic fit has been used in each segment. The detrended series $\overline{Y}$ in each segment is then the difference between Y and the quadratic fit.

Given these detrended segments $\overline{Y}_{(j)}$, ~~the formula in~~ equation 1.8 ~~gives~~ provides the s-dependent fluctuation coefficient $F^{(n)}(s)$:

$$F^{(n)}(s) = \sqrt{\frac{1}{\lfloor N/s \rfloor} \sum_{j=0}^{\lfloor N/s \rfloor - 1} \mathrm{Var}(\overline{Y}_{(j)})}. \tag{1.8}$$

We define the DFA exponent in relation to the fluctuation coefficient as follows:

Definition 1.1.1 (The DFA scaling exponent) *The order-n DFA scaling exponent α_n is the power law decay rate of the fluctuations coefficient $F^{(n)}(s)$ with decreasing s, that is,*

$$F^{(n)}(s) \sim s^{\alpha_n}. \tag{1.9}$$

Practically, $F^{(n)}(s)$ is calculated for many values of s and the value of α is found to be the slope of the linear best fit line to the logarithmic plot of over s. Figure 1.1c shows the order-2 DFA coefficient $F^{(2)}(s)$ of the red noise time series in figure 1.1a for $s = 1, 2, \ldots, 100$. The slope of the logarithmic plot in figure 1.1c , obtained using a simple linear fit, is the DFA exponent α. In this study we consistently use an order-2 DFA method, a quadratic fit during the detrending step of DFA. Following the approach of [Livina and Lenton, 2007] we measure the DFA exponent in the temporal range $10 \leq s \leq 100$.

Kantelhardt et al. [2001] also proposes a modified DFA method designed to provide better results with shorter time series. long-range correlations can cause an overestimate of the exponent α for short time series. The modified method therefore involves multiplication of the fluctuation coefficient $F^{(n)}(s)$ by a correction function derived from $F^{(n)}(s)$ but with the segments Y shuffled in a random order to destroy any long-range correlations in the data.

1.1.3 The power spectrum exponent

The power spectrum scaling exponent β is calculated by estimating the slope of the power spectrum $S(f)$ of the data, plotted on logarithmic axes [Bak et al., 1988], from which one can obtain β via the scaling relationship

$$S(f) \sim f^{-\beta}, \tag{1.10}$$

where f is frequency and S is power ...

In this study we have calculated the power spectrum scaling exponent using the standard periodogram, obtained from the absolute value of the fast Fourier transform. That is, given time series X we take the FFTW3 fast Fourier transform [Frigo and Johnson, 1998, 2005] (implemented in Matlab) and plot this on logarithmic axes. In figure 1.1d this is done for the red noise series shown in figure 1.1a. The slope is measured inside the frequency range $10^{-2} \leq f \leq 10^{-1}$ (dashed vertical line) corresponding to the time range of 10 to 100 units used by Livina and Lenton [2007] in the measurement of the DFA exponent, and to put the technique in agreement with the ACF, and DFA methods described above 1.1b,c.

This time / frequency range represents short-term memory in the data, which is sensitive to onset of transitions and bifurcations, thus providing early warning signals of tipping.

Table 1.1 Table showing the mean values of three scaling indicators, and lag-1 ACF, applied to different noise series.

	Exponent			
Noise species	DFA (α)	PS (β)	ACF (γ)	lag-1 ACF
White	0.510	0.0110	-0.0163	-0.00371
~~Red~~ Brown	1.50	2.04	0.473	0.993
Red 63	1.48	1.92	0.565	0.990

There exist various methods for the calculation of the power spectrum besides the standard periodogram used here [Bingham et al., 1967], many of which are developed for specific applications in engineering. Two of the most common alternatives, Welch's method [Welch, 1967] and the Multitaper method [Thomson, 1982], were ~~used~~ tested experimentally but did not appear to provide any additional benefit for the purposes of this study.

1.2 Relationships between scaling exponents

The three scaling exponents defined in section 1.1 are illustrated in figure 1.1 where they are applied to a red noise series. The red noise, in this case, is long-range correlated, generated using the method described by Kasdin [1995] (see equation 1.3). Since all three methods give a measure of time-scaling, there is a relationship between them. We calculate the value of each exponent for three different species of noise: white noise is produced by sampling each point from a normal $N(0,1)$ distribution; ~~simple red~~ Brown noise is the cumulative sum of such a white noise series; long-range correlated red noise is generated according to Kasdin [1995] (see equation 1.3) using an AR(63) model with parameter $\lambda = 2$. The results are shown in table 1.1, in each case the value given is the mean value of the exponent calculated for 100 series of length 10000. The long-range correlated red noise is labelled "Red 63". The simple lag-1 ACF is also included for comparison.

It is random walk

The values of α and β in table 1.1 confirm the linear relationship

$$\alpha = \frac{1+\beta}{2} \tag{1.11}$$

noted by Kantelhardt et al. [2001] to apply generally to these exponents. The relationship is also noted by Heneghan and McDarby [2000] to exist only for long-range correlated noise series, although the result here is the same for the simple red noise, an AR(1) process, and the AR(63) process. In a further experiment, the DFA and PS exponents are calculated for multiple noise series with spectral densities between white and red noise. 200 short-range correlated noise series are generated using the AR(1) model

What length?

$$z_n = \lambda z_{n-1} + \eta_n \quad \lambda = 0, 0.005, \ldots, 1 \tag{1.12}$$

where η is white noise. The DFA and PS exponents are calculated for each series. 200 further, long-range correlated, series are generated using the AR(63) model [Kasdin, 1995] (equation 1.3) with parameter λ between 0 and 2. The results are shown in figure 1.3. We see that although the linear relationship given by equation 1.11 is confirmed by the long-range correlated noise (panel **b**), but not exactly by the short-range (panel **a**).

~~In addition to the two species of noise noted above, a different species was used in initial experiments in place of the long-range correlated noise generated using the AR(63) model implemented by the MatLab dsp systems toolbox [Kasdin, 1995]~~. The alternate method of producing long range correlated noise was to transform the power spectrum of white noise in the frequency domain, thus changing the PS exponent β [Makse et al., 1996]. That is, we first sample from a Gaussian distribution to approximate white noise, $X(t)$, we then take the fast Fourier transform of this white noise series (denoted $\hat{X}(f)$) and then transform this by

$$\hat{Y}(f) := \hat{X}(f)\sqrt{f^{-\beta}}. \tag{1.13}$$

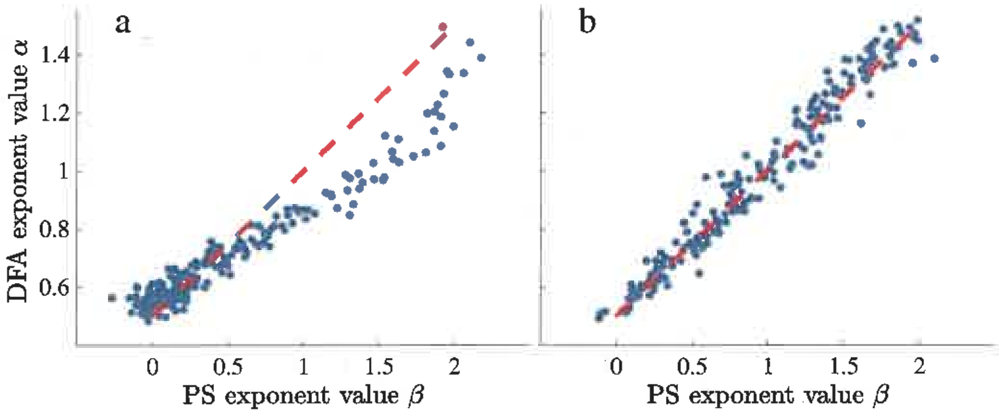

Fig. 1.3 ~~Showing the~~ DFA exponent α plotted against the PS exponent β for short-range correlated (panel **a**) and long-range correlated (panel **b**) noise with varying correlation parameters. The result for each noise series is represented by one blue dot, the expected linear relationship is shown in red, see equation 1.11. Length of data?

Finally, we use the inverse fast Fourier transform to obtain the series $Y(t)$ in the time domain, which has power spectrum $S(f)$ proportional to $f^{-\beta}$. The result when using this method of noise generation is quantitatively and qualitatively similar to the result when using the AR(63) model (figure 1.3 panel **b**): the DFA and PS exponents are strongly correlated and obey the linear relationship $\alpha = (1+\beta)/2$. The advantage of this method is control of exponent β in time series

Kantelhardt et al. [2001] notes that a direct calculation of the correlations exponent, that is the ACF scaling exponent γ, is often unsuitable due to noise and underlying trends. The DFA method is therefore introduced as an indirect measure of the exponent. In our own implementations of the two methods, the ACF exponent is less reliable. Figure 1.4 illustrates this in the case that both methods are applied to noise series generated using the AR(63) method [Kasdin, 1995] with varying values of the parameter λ. Fifty noise series are generated for each value $\lambda = 0, 0.02, 0.04, \ldots, 1$, at each value λ the scaling exponent is calculated for each of the 50 series and the standard deviation over these 50 exponents is found. We see that the DFA exponent is consistent in that the standard deviation of its value is low, and indeed the standard deviation is, itself, consistently low. The standard deviation is much larger in the ACF exponent, although the exponents have similar mean values (between 0 and 1).

What length?

bifurcation with a decaying stable mode at $z = 0$. For $\mu < 0$ this mode becomes unstable and two stable modes are created at $z = \pm\sqrt{-\mu/2}$. The shape of the potential function U is illustrated in figure 1.6. Differentiating U, we rewrite equation 1.16 as

$$\frac{dz}{dt} = -4z^3 - 2\mu z = f(z). \tag{1.17}$$

We now consider the $\mu > 0$ system with a small perturbation from the stable mode, that is, $z = \varepsilon$. Linearising equation 1.17 using the first-order Taylor expansion around the stable mode $z = 0$ gives

$$\frac{d}{dt}(\varepsilon) = f(\varepsilon) \approx -2\mu\varepsilon, \tag{1.18}$$

so, for $\mu > 0$, the system goes exponentially to zero. When $\mu = 0$ the recovery rate -2μ is zero and so the system will not recover from the small perturbation ε. When $\mu < 0$ the mode $z = 0$, as noted, is unstable and system will go exponentially away from zero. If it is assumed that a dynamical system possesses a critical mode, that is, a mode that is decaying as a result of a bifurcation, then the recovery rate of this mode will go to zero as the system takes longer (eventually infinite time, after the bifurcation) to return to the stable state after a perturbation. The tendency of the system in one direction can be measured by an increase in the lag-1 autocorrelation function (ACF1), thus this is a reasonable indicator of the decreasing decay rate.

The hypothesis is that this critical slowing down will be seen in a wide variety of dynamical systems approaching a bifurcation. Van Nes and Scheffer [2007] test this hypothesis with a number of ecological models and find that the intensity of the critical slowing down effect is linearly, or almost linearly, related to the distance from the tipping point in all cases.

Held and Kleinen [2004] use a rise in ACF1 as an indicator of a bifurcation in a model of North Atlantic thermohaline circulation. The measured variables are presumed to represent a dynamical system undergoing a bifurcation and therefore possessing a critical mode. In order

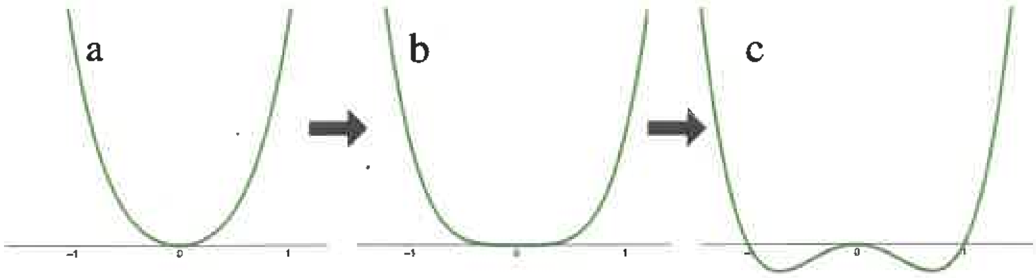

Fig. 1.6 Illustrating the changing shape of the potential function U in equation 1.16. U is shown for $\mu > 0$ where the shape is a "single well" (a); for $\mu = 0$ where the bifurcation occurs; and for $\mu < 0$ when the "double well" shape has developed.

to determine the point at which the ACF1 value rises, a method is used labelled "degenerate fingerprinting" by Held and Kleinen [2004]. The ACF1 of the data, in this case the Atlantic overturning flux, is calculated in sequential, overlapping windows so that the ACF1 value can be tracked over time. If the value is expected to rise, or begin to rise, before the tipping point, the increase in the ACF1 is an early warning signal of the tipping.

In this study we refer to this time-dependent ACF1 as *the ACF1 indicator*, giving us the following definition:

Definition 1.3.1 (The ACF1 indicator) *For time series data*

$$[x_t] = x_1, x_2, x_3, ..., x_N \tag{1.19}$$

we create a new indicator series $[Y_t]$ ~~*where*~~

$$Y_t = ACF_1([x_{t-w+1}, ..., x_t]) \;\; for \;\; t \geq w \tag{1.20}$$

where w is the window size, typically $\leq 10\%$ *of the time series length N. The first* $w-1$ *values of* $[Y_t]$ *are simply set as equal to zero* or omitted.

This method is used throughout this study and, particularly when observed physical data is used in chapter 3, careful attention is paid to the choice of the window size w.

Livina and Lenton [2007] modify the ACF1 indicator so that the DFA exponent is calculated instead of the lag-1 autocorrelation function in each window of data:

Definition 1.3.2 (The DFA indicator) *For time series data $[x_t] = x_1, \ldots, x_N$ we create the DFA indicator series $[Y_t]$, where*

$$Y_t = DFA([x_{t-w+1}, \ldots, x_t]) \ \textit{for} \ \ t \geq w, \tag{1.21}$$

where w is the window size, typically $\leq 10\%$ of the time series length N. The function $DFA(\cdot)$ denotes the calculation of the order-2 DFA exponent α (definition 1.1.1). $Y_t = 0$ for $t < w$.

In this study, we consistently use the order-2 DFA exponent, but the indicator may also be used with higher orders. Livina and Lenton [2007] test the DFA indicator method on data from a model of Atlantic overturning circulation [Edwards and Marsh, 2005] and also Greenland paleotemperature records [Alley, 2000]. The DFA indicator is also compared to the ACF1 indicator [Lenton et al., 2012] where it is concluded that both methods offer their own pros and cons. In particular, computing the DFA exponent for every window of data is ~~very~~ computationally expensive.

The possibility of such modification of the fingerprinting method allows us to instead use the power spectrum (PS) scaling exponent, which is obtained from the periodogram as in section 1.1.3, [Prettyman et al., 2018]. Kleinen et al. [2003] notes already that spectral properties of the time series may also be used as tipping point indicators, in particular when the tipping point is associated with a change in the structure of the noise, or the stationary system becoming non-stationary. A shift from a dominance of short-scale memory to long-scale memory, seen in the measurement of the PS scaling exponent, will be associated with a rise in autocorrelation. In this respect, therefore, the PS exponent and the ACF are related, further justifying the use of the PS exponent as an EWS. There also exists a relationship between the PS exponent and the DFA exponent [Heneghan and McDarby, 2000; Kantelhardt et al., 2001] as explained in section 1.2. We note that although Heneghan and McDarby [2000] advises that there is no difference between the DFA and PS exponents, and that therefore it is not necessary to use both, this applies only for long-range correlated data. The relationship is investigated in section 1.2: while figure 1.3b shows the linear relationship proposed by Kantelhardt et al. [2001], figure 1.3a shows a non-linear relationship in the experiment where short-range correlated noise is used. We thus see that the relationship between the two exponents may be vary depending on the nature of the data. In a system where the correlation in the time series changes over time, (and trends or feedbacks affect the dynamical system) there may not exist an analytic relationship between the DFA and PS exponents and the use of one may therefore reveal information that the use of the other cannot.

We therefore propose the use of the power spectrum indicator, alongside the DFA and ACF1 indicators. The PS exponent is estimated by calculating the negative slope of the periodogram, produced using a fast Fourier transform, in the frequency range $[10^{-1}, 10^{-2}]$. This is done for in a sliding window according to the fingerprinting method [Held and Kleinen, 2004; Livina ~~et al., 2012~~ and Lenton, 2007].

Definition 1.3.3 (The PS indicator) *For time series data $[x_t] = x_1, \ldots, x_N$ we create the PS indicator series $[Y_t]$ ~~where~~*

$$Y_t = PS([x_{t-w+1}, \ldots, x_t]) \;\; for \;\; t \geq w \tag{1.22}$$

where w is the window size and $Y_t = 0$ for $t < w$. The function $PS(\cdot)$ denotes, in this instance, the calculation of the power spectrum scaling exponent β as the negative slope of the periodogram (equation 1.10).

1.4 Comparison of indicators

In this section we apply the DFA, PS and ACF1 indicators to two examples of time series. One is constructed from segments of generated noise, artificially designed to have a gradually

ACF for noise series with different values of β (figure 1.5), the lag-1 ACF increses linearly with increasing β only up to a point, approximately $\beta = 1$, after which the rate of increase slows. The DFA and PS increase linearly, as expected, with increasing β and the relationship between the two indicators obeys the same linear law connecting to DFA and PS exponents (equations 1.11). The variance in both indicator series is also similar, and so we have evidence for the assertion that it is not necessary to use the PS exponent when the DFA exponent is available [Heneghan and McDarby, 2000], but concerning the sliding-window indicator methods, rather than a simple calculation of the exponents.

1.4.2 Early warning signals in a pitchfork bifurcation model

We now apply the indicators to a time series produced by a model of a bifurcating dynamical system. The system is the potential-well system given by the stochastic differential equation

$$\dot{z}(t) = -\frac{\partial}{\partial z}U + \sigma\eta_t \tag{1.23}$$

where η_t is a white noise series with mean 1, $\sigma = 0.05$ and U is the potential function given by

$$U = z^4 + \left(3 - \frac{t}{200}\right)z^2. \tag{1.24}$$

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V The shape of the potential function U changes with increasing t in the same ~~same~~ at the potential function in equation 1.16 changes with decreasing μ and is illustrated in figure 1.6. In the range $t < 600$, U has a "single well" shape such that the system has a single stable node $z = 0$. As t increases the base of the well becomes shallower, thus the system takes longer to return to the stable node $z = 0$ after a perturbation and the variance around this point increases. At $t = 600$ the system experiences a bifurcation when the second derivative of U becomes zero. At this point the shape of U evolves into a "double well" and node $z = 0$ becomes unstable whilst two ?

stable nodes develop at

$$z = \pm\frac{\sqrt{t-600}}{200}. \tag{1.25}$$

Equation 1.23 is integrated numerically with a time step of $\delta t = 0.05$, from $t = 1$ to $t = 1000$, and sampled with $\Delta t = 1$ so that the resulting time series has length 1000. The ACF1, DFA and PS indicators are then applied with a window size of 100 points. This experiment is repeated for ~~100 times, that is,~~ 100 such time series of length 1000 ~~are produced~~ and the indicators are applied to all of them. The means of the indicator series are found and the results are shown in figure 1.8. We see that all three indicators provide an early warning signal: they show an increasing trend before the bifurcation occurs, already visible (in the mean) at $t = 500$. The trend is most obvious in the ACF1 indicator, the variances of the DFA and PS indicators are much larger so that the increase in the PS indicator may not be detectable above the level of one std.

The experiment demonstrated the agreement between indicators.

1.5 Discussion

We have developed a novel indicator of EWS for tipping points in dynamical systems by estimating the power-law decay rate of the power spectrum in a sliding window [Prettyman et al., 2018]. We have applied this PS indicator to an artificial time series with an increasing scaling exponent, and a model including a bifurcation.

We have found that the PS indicator behaves similarly to the previously used ACF1 and DFA indicators applied to the model data, but with larger variance, which limits its usefulness in situations where the model has a noisy trajectory or cannot be observed multiple times.

Of the ACF, DFA and PS scaling exponents, all may be used as EWS indicators, although Kantelhardt et al. [2001] notes that a direct calculation the ACF exponent γ is often unsuitable and the DFA method is therefore introduced as an indirect measure. In our own implementations of the two methods, the ACF exponent is less reliable. The less computationally expensive lag-1 autocorrelation function is also often used instead of the ACF exponent since both provide a measure of critical slowing down (see equation 1.14). But it is highly sensitive to trends and nonstationarity

Heneghan and McDarby [2000] notes that the DFA and PS exponents have a predictable relationship and therefore the ~~use of both is not necessary~~ choice between them depends on the system.